In [ ]:
```
# Install the official UCI ML Repository package
!pip install ucimlrepo

from ucimlrepo import fetch_ucirepo

# Fetch the Seoul Bike Sharing Demand dataset (ID: 560)
seoul_bike_sharing_demand = fetch_ucirepo(id=560)

# Extract the original dataframe
df_raw = seoul_bike_sharing_demand.data.original

# Save it as a CSV file in your Colab files
df_raw.to_csv('SeoulBikeData.csv', index=False)

print("SeoulBikeData.csv has been successfully generated and saved!")
```

```
Collecting ucimlrepo
  Downloading ucimlrepo-0.0.7-py3-none-any.whl.metadata (5.5 kB)
Requirement already satisfied: pandas⩾1.0.0 in /usr/local/lib/python3.13/dist-
packages (from ucimlrepo) (2.2.3)
Requirement already satisfied: certifi⩾2020.12.5 in
/usr/local/lib/python3.13/dist-packages (from ucimlrepo) (2026.7.22)
Requirement already satisfied: numpy⩾1.26.0 in /usr/local/lib/python3.13/dist-
packages (from pandas⩾1.0.0→ucimlrepo) (2.1.3)
Requirement already satisfied: python-dateutil⩾2.8.2 in
/usr/local/lib/python3.13/dist-packages (from pandas⩾1.0.0→ucimlrepo)
(2.9.0.post0)
Requirement already satisfied: pytz⩾2020.1 in /usr/local/lib/python3.13/dist-
packages (from pandas⩾1.0.0→ucimlrepo) (2025.2)
Requirement already satisfied: tzdata⩾2022.7 in
/usr/local/lib/python3.13/dist-packages (from pandas⩾1.0.0→ucimlrepo)
(2026.4)
Requirement already satisfied: six⩾1.5 in /usr/local/lib/python3.13/dist-
packages (from python-dateutil⩾2.8.2→pandas⩾1.0.0→ucimlrepo) (1.17.0)
Downloading ucimlrepo-0.0.7-py3-none-any.whl (8.0 kB)
Installing collected packages: ucimlrepo
Successfully installed ucimlrepo-0.0.7
SeoulBikeData.csv has been successfully generated and saved!
```

In [ ]:
```
!pip install ucimlrepo

import pandas as pd
import numpy as np
import statsmodels.formula.api as smf
import statsmodels.api as sm
import seaborn as sns
import matplotlib.pyplot as plt
from ucimlrepo import fetch_ucirepo
```

```
# Generate the CSV file directly from the UCI repository
seoul_bike_sharing_demand = fetch_ucirepo(id=560)
df_raw = seoul_bike_sharing_demand.data.original
df_raw.to_csv('SeoulBikeData.csv', index=False)
print("SeoulBikeData.csv successfully generated!\n")

# Part A: Load and clean data
df = pd.read_csv('SeoulBikeData.csv')

# Explicitly rename all 14 columns to safe, symbol-free names
df.columns = [
    'Date', 'Rented_Bike_Count', 'Hour', 'Temperature', 'Humidity',
    'Wind_speed', 'Visibility', 'Dew_point_temp', 'Solar_Radiation',
    'Rainfall', 'Snowfall', 'Seasons', 'Holiday', 'Functioning_Day'
]

# Part B: EDA (Scatterplot)
sns.scatterplot(x='Temperature', y='Rented_Bike_Count', data=df, alpha=0.3)
plt.title('Bike Demand vs Temperature')
plt.show()

# Part C: Simple Linear Regression
slr = smf.ols('Rented_Bike_Count ~ Temperature', data=df).fit()
print("--- Simple Linear Regression ---")
print(slr.summary())

# Part D, E, G: Multiple Regression with Dummy Variables & Inference
mlr = smf.ols('Rented_Bike_Count ~ Temperature + Humidity + C(Seasons)',
data=df).fit()
print("\n--- Multiple Linear Regression ---")
print(mlr.summary())

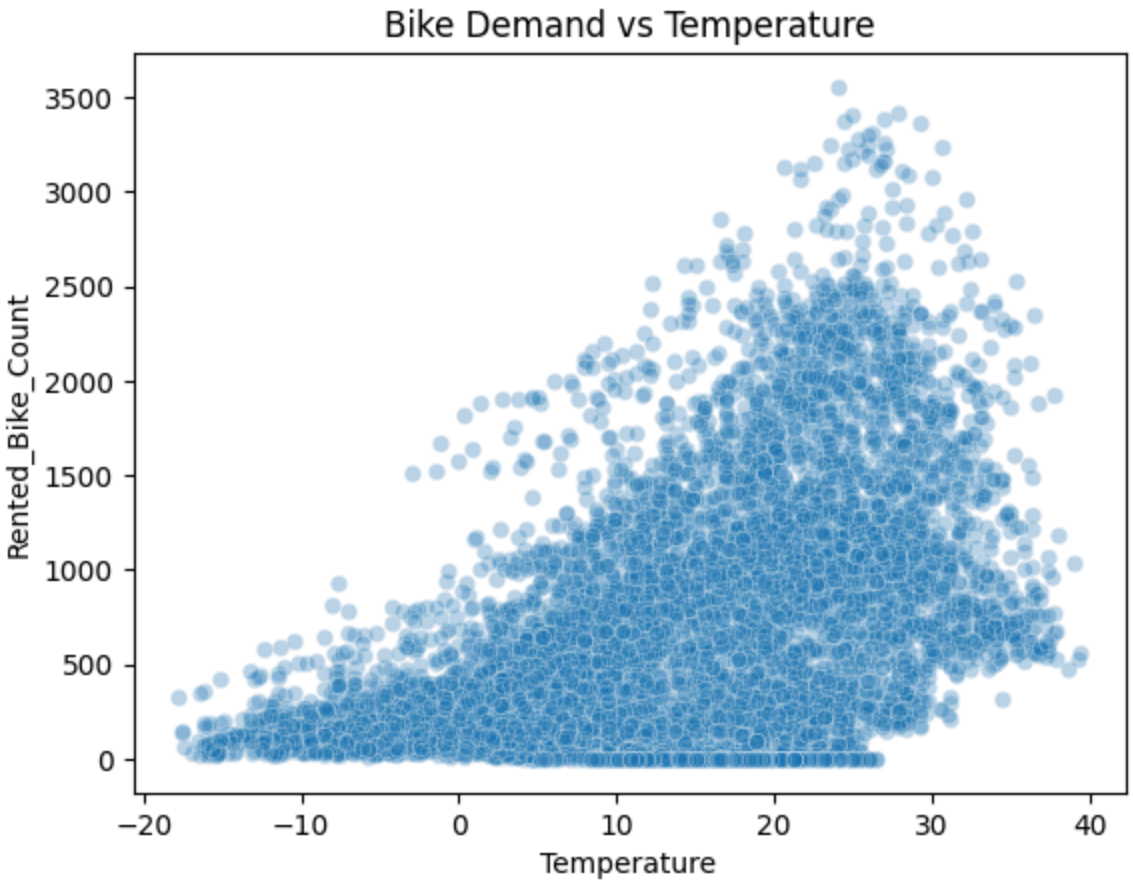
# Part F: Matrix Form Demonstration (Using a small subset of 5 rows)
subset = df.head(5)
X = sm.add_constant(subset[['Temperature', 'Humidity']])
y = subset['Rented_Bike_Count']
X_mat = np.array(X)
y_mat = np.array(y)
beta_hat = np.linalg.inv(X_mat.T @ X_mat) @ X_mat.T @ y_mat
print("\n--- Matrix Beta Hat (Intercept, Temp, Humidity) ---")
print(beta_hat)

# Part H: Two-Way ANOVA & Interactions
interaction_model = smf.ols('Rented_Bike_Count ~ Temperature * C(Seasons)',
data=df).fit()
print("\n--- Two-Way ANOVA ---")
print(sm.stats.anova_lm(interaction_model, typ=2))

# Part I: Diagnostics (Residual Plot)
```

```
sns.residplot(x=interaction_model.fittedvalues, y=interaction_model.resid,
lowess=True)
plt.title('Residuals vs Fitted (Constant Variance Check)')
plt.show()
```

```
Requirement already satisfied: ucimlrepo in /usr/local/lib/python3.13/dist-
packages (0.0.7)
Requirement already satisfied: pandas⩾1.0.0 in /usr/local/lib/python3.13/dist-
packages (from ucimlrepo) (2.2.3)
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Requirement already satisfied: python-dateutil⩾2.8.2 in
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(2.9.0.post0)
Requirement already satisfied: pytz⩾2020.1 in /usr/local/lib/python3.13/dist-
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Requirement already satisfied: tzdata⩾2022.7 in
/usr/local/lib/python3.13/dist-packages (from pandas⩾1.0.0→ucimlrepo)
(2026.4)
Requirement already satisfied: six⩾1.5 in /usr/local/lib/python3.13/dist-
packages (from python-dateutil⩾2.8.2→pandas⩾1.0.0→ucimlrepo) (1.17.0)
SeoulBikeData.csv successfully generated!
```



```
--- Simple Linear Regression ---
                            OLS Regression Results
==============================================================================
Dep. Variable:      Rented_Bike_Count   R-squared:                       0.290
Model:                            OLS   Adj. R-squared:                  0.290
Method:                 Least Squares   F-statistic:                     3578.
Date:                Tue, 13 Oct 2026   Prob (F-statistic):               0.00
Time:                        12:01:59   Log-Likelihood:                -67600.
No. Observations:                8760   AIC:                         1.352e+05
Df Residuals:                    8758   BIC:                         1.352e+05
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     329.9525      8.541     38.631      0.000     313.210     346.695
Temperature    29.0811      0.486     59.816      0.000      28.128      30.034
==============================================================================
Omnibus:                      954.681   Durbin-Watson:                   0.271
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             1421.965
Skew:                           0.817   Prob(JB):                    1.68e-309
Kurtosis:                       4.108   Cond. No.                         25.9
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is
correctly specified.

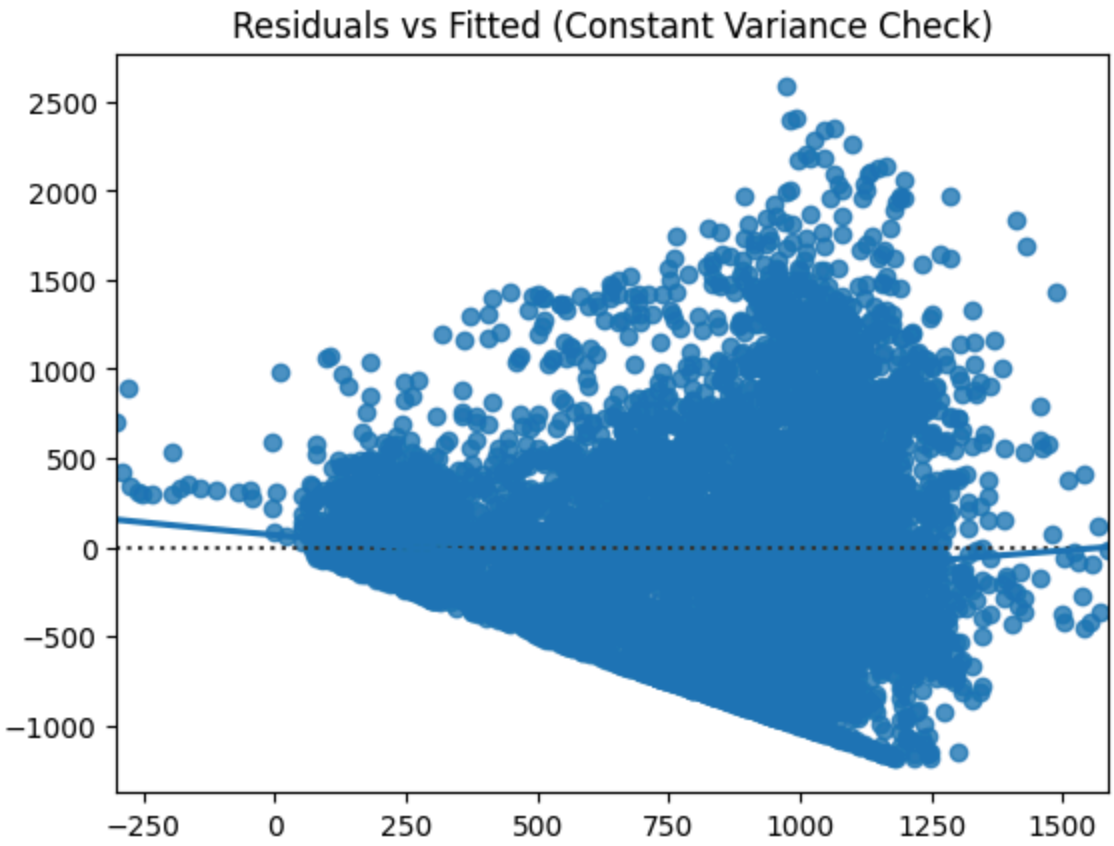
--- Multiple Linear Regression ---
                            OLS Regression Results
==============================================================================
Dep. Variable:      Rented_Bike_Count   R-squared:                       0.384
Model:                            OLS   Adj. R-squared:                  0.384
Method:                 Least Squares   F-statistic:                     1092.
Date:                Tue, 13 Oct 2026   Prob (F-statistic):               0.00
Time:                        12:01:59   Log-Likelihood:                -66976.
No. Observations:                8760   AIC:                         1.340e+05
Df Residuals:                    8754   BIC:                         1.340e+05
Df Model:                           5
Covariance Type:            nonrobust
==============================================================================
==========
                             coef    std err          t      P>|t|      [0.025
0.975]
------------------------------------------------------------------------------
---------
Intercept             1013.7301     24.804     40.870      0.000     965.109
1062.351
C(Seasons)[T.Spring]   -64.8196     15.312     -4.233      0.000     -94.835
-34.804
```

```
C(Seasons)[T.Summer]   -67.6482     19.213     -3.521      0.000    -105.310
-29.986
C(Seasons)[T.Winter]  -234.3338     21.980    -10.661      0.000    -277.421
-191.247
Temperature             27.1388      0.907     29.912      0.000      25.360
28.917
Humidity                -9.7479      0.279    -34.985      0.000     -10.294
-9.202
==============================================================================
Omnibus:                     1168.742   Durbin-Watson:                   0.322
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             2073.796
Skew:                           0.879   Prob(JB):                         0.00
Kurtosis:                       4.610   Cond. No.                         370.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is
correctly specified.

--- Matrix Beta Hat (Intercept, Temp, Humidity) ---
[478.27352941 186.61764706  20.09411765]

--- Two-Way ANOVA ---
                             sum_sq      df            F         PR(>F)
C(Seasons)             2.933409e+07     3.0    34.453476   3.961081e-22
Temperature            3.205296e+08     1.0  1129.405158  5.450030e-233
Temperature:C(Seasons) 7.384386e+07     3.0    86.731081   2.680765e-55
Residual               2.483852e+09  8752.0          NaN            NaN
```



Summative Assessment 1: Comprehensive Linear Modeling Analysis

Introduction and Research Problem

Urban bike-sharing systems provide crucial micro-mobility solutions, but maintaining optimal inventory across stations requires accurate demand forecasting. The research problem is to identify how meteorological and seasonal factors drive the hourly utilization of Seoul's public bike system.

Dataset Description

In this analysis, we used the Seoul Bike Sharing Demand Dataset taken from the UCI Machine Learning Repository.

The response variable (Y) would be the number of rented bikes, which is measured by the absolute count in one hour.

The explanatory variables (X) consist of temperature in degrees Celsius (°C), humidity in percentage (%), and seasons (Spring, Summer, Autumn, Winter). Data cleaning was done to check for missing values. The columns were renamed to enable correct computation.

Research Questions & Hypotheses

Question: How do temperature, humidity, and seasonality interact to predict hourly bike-sharing demand?

Hypotheses:H_0: Environmental factors have no linear effect on bike demand ($\beta_1 = \beta_2 = ... = 0$). H_a: Environmental factors significantly predict bike demand.

Exploratory Data Analysis

The scatterplot of Bike Demand versus Temperature reveals a distinct positive correlation; as temperature rises, bike rentals increase. However, the variance in rentals widens significantly at higher temperatures, indicating potential heteroscedasticity.

Simple Linear Regression

Equation:

$$\text{Rented_Bike_Count} = 329.9525 + 29.0811 \cdot \text{Temperature}$$

Interpretation: For every 1°C increase in temperature, hourly bike demand increases by roughly 29.08 bikes, on average.

Model Fit: The model yields an R^2 of 0.290, meaning temperature alone explains 29.0% of the variance in bike rentals.

Multiple Linear Regression

To control for confounding variables, the model was expanded:Equation:

$$\text{Rented_Bike_Count} = 1013.7301 + 27.1388\text{Temp} - 9.7479\text{Humidity} - 64.8196\text{Spring} - 67.6482\text{Summer}$$

Interpretation: Holding seasonality and temperature constant, a 1% increase in humidity is associated with a decrease of 9.75 rented bikes.

Evaluation: The adjusted R^2 increased to 0.384, confirming that adding humidity and seasons improved the model's predictive power without overfitting.

Statistical Inference

Hypothesis Tests: Since the p-values for the Temperature and Humidity variables are both equal to 0.000, it indicates that the null hypothesis can be rejected in both cases. Hence both variables can be viewed as significant predictors.

Confidence Intervals: A 95% confidence interval for the actual slope of the population in reference to Temperature is between 25.360 and 28.917.

Practical Significance: These results are statistically significant but the implication of a change of 1% in humidity (-9.75 bikes) is negligible compared to the effect of the change in seasons like, for example, -234.33 bikes in Winter.

Matrix Formulation and Least-Squares Demonstration

The multiple regression model operates on the matrix equation $y = X\beta + \epsilon$:

y: The column vector of observed Rented Bike Counts.

X: The design matrix containing a column of 1s (for the intercept) and columns for the observed predictor values.

β: The unobserved population parameters.

ϵ: The vector of random error terms.

The Python matrix calculation yields a Beta Hat of [478.27, 186.61, 20.09] for the 5-row subset, proving that software solves for the least-squares estimates $\hat{\beta}$ by minimizing the sum of squared residuals algebraically.

Dummy Variables & 10. ANOVA

Reference Category: The software alphabetically selected Autumn as the reference category.

The interpretation of the coefficient associated with the Winter dummy variable (-234.3338) indicates that, with constant temperature and humidity, bike rental activities during the season of Winter would be lower than during the season of Autumn by around 234 rentals per hour.

The association with ANOVA is that both ANOVA and regression using dummy variables are mathematically equivalent in so far as they decompose the variance in bike rentals data to check whether the mean belonging to the factor of season are statistically different from the reference level

Interaction Analysis

An interaction term between Temperature and Seasons was introduced to test if the slope of temperature changes depending on the time of year.

Interpretation: The significant interaction p-value (2.68×10^{-55}) proves that the effect of temperature is not constant. An increase of $1^{\circ}\mathrm{C}$ drives a different rate of increase in bike rentals depending on the season.

Model Diagnostics

The presumption of constant variance has been violated in this case (homoscedasticity). The "Residuals vs. Fitted" plot clearly depicts a cone shape; as expected, the variance of the residuals increases with the increase in demand for bicycles. Since count data inherently has a variance that grows with the mean, the next step for further analysis will be to carry out a logarithmic transformation of the response variable or a Poisson regression.

Final Model & 14. Conclusions

The best fit model is Multiple Linear Regression model that includes the Temperature $\times$ Season interaction. The results suggest that external factors have great influence on the mobility patterns. Generally, bike demand in Seoul can be predicted based on temperature, however, one should keep in mind the seasonal effects of temperature on mobility.

References

Seoul Bike Sharing Demand Dataset. UCI Machine Learning Repository (ID: 560).
https://archive.ics.uci.edu/dataset/560/seoul+bike+sharing+demand